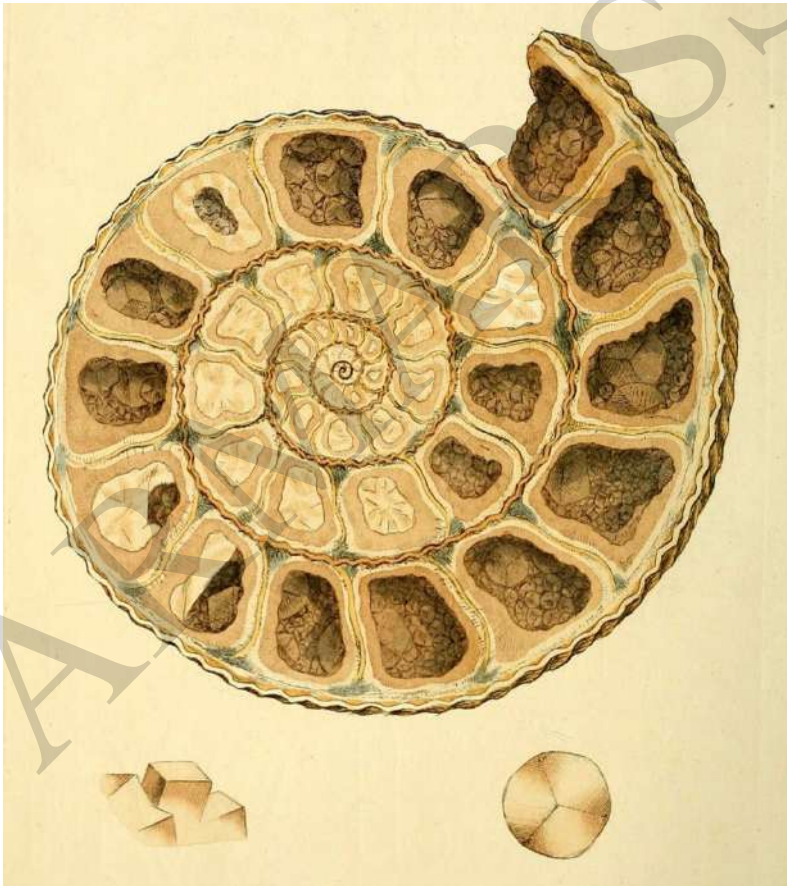
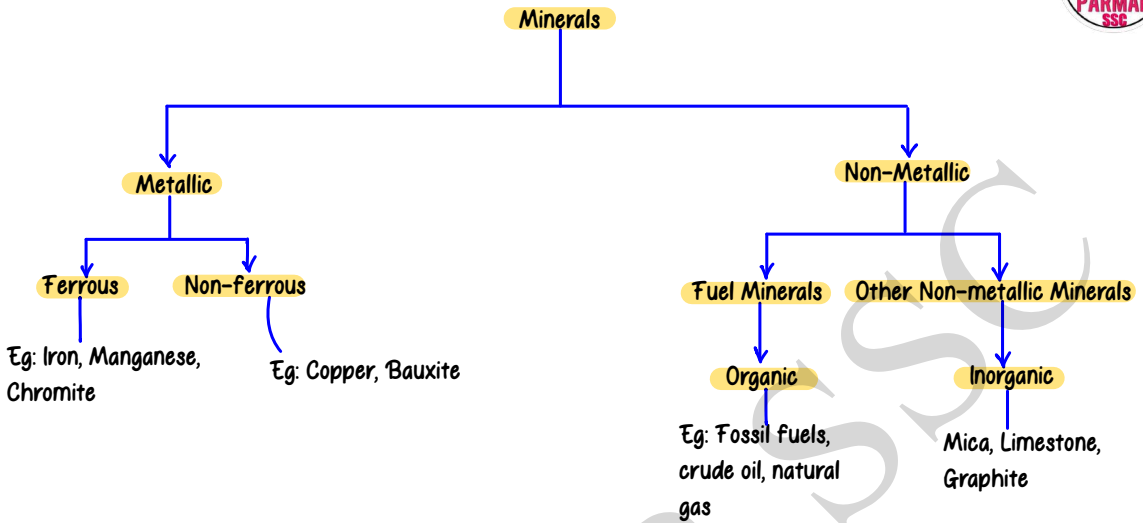


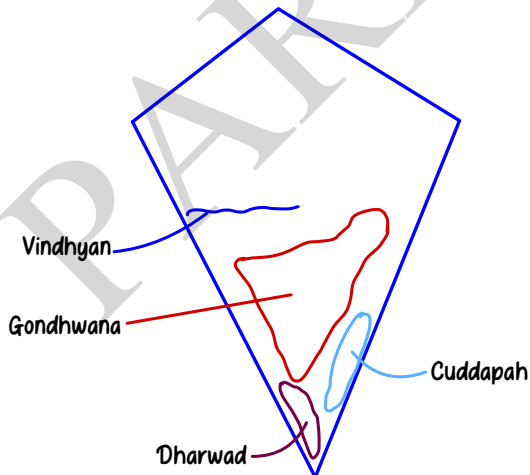
MINERALS



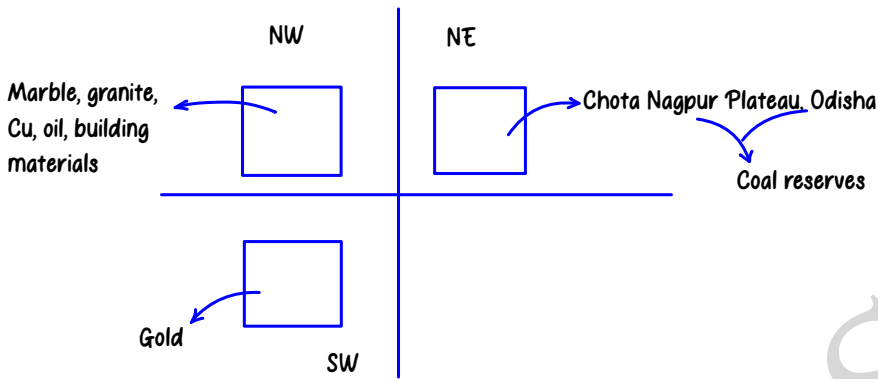


Mining → Ore = Metal + Impurities
 ↓
 Refining → Gives metals/non-metals

Distribution of Minerals



- **Gondwana:** Gold
- **Dharwad/Cuddapah:** Gold, Iron
- **Vindhyan:** Non-metallic minerals



Iron

- Ores
 - Hematite
 - Magnetite → Has highest % of iron (roughly 70%)
 - Limonite

Reserves

- **Topmost:** Odisha (Hematite)
- Karnataka (Magnetite)
- **Topmost reserve in the world:** Australia

Production

- Odisha > Chattisgarh > Karnataka



State

- **Odisha**
- **Jharkhand**
- **Chhattisgarh**

Region

Sundargarh
Mayurbhanj
Jhar

Singhbhum

Durg
Bastar

Mines

Badampahar

Noamundi

Dalli Rajhara
Bailadila
Dantewada

• **Karnataka**

Ballari
Chikkamangaluru
Chitradurga

Baba budan giri
Kudremukh

Manganese (Mn)

Reserve

• **Topmost in world:** South Africa

• **Topmost in India:** Odisha

- Sundargarh
- Keonjhar
- Kalahandi
- Mayurbhanj
- Koraput

• **Karnataka:** Ballari, Chikkamangaluru

Bauxite: Ore of Aluminium; non-ferrous

Reserve

• **Topmost in world:** Australia

• **Topmost in India:** Odisha

• **Jharkhand:** Lohardaga, Palamu, Ranchi

• **Gujarat:** Bhavnagar, Junagarh

• **Madhya Pradesh:** Katni, Balaghat

Few Aluminium producing Companies and their HQ

• **NALCO** → HQ: Bhubaneswar (Odisha)

• **BALCO** → HQ: New Delhi

• **HINDALCO** → HQ: Mumbai

• **Vedanta Resources** → HQ: Mumbai

Copper

Production

• **Topmost in India:** Madhya Pradesh

- Balaghat (mines)
- Malajkhand (region)

• Rajasthan

- Jhunjhunu and Alwar (regions)
- Khetri (mines)

• Jharkhand: Singhbhum

Mica: it is used as insulators in electronics and electrical appliances

Production

- India: Andhra (top), J & K, Bihar

Gold

- Kolar Gold Fields
 - Hutti Gold Mines (Raichur)
 - Ramagiri: Andhra Pradesh
- } → Karnataka

Diamond

- India: Madhya Pradesh (Panna) is the only state that produces diamond
- Topmost in World: Russia

Lithium: known as "White Gold"

World

- Reserve (top): Chile
- Production (top): Australia

Coal

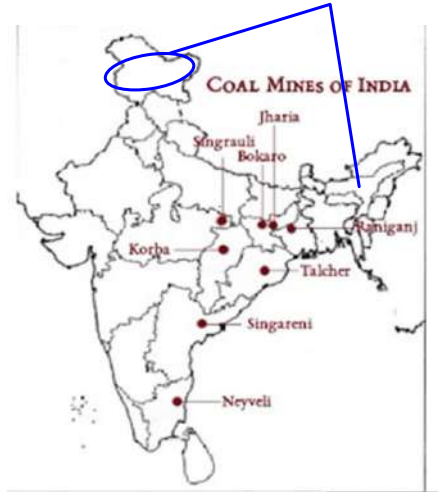
- Known as "Black Gold" and "Buried Sunshine"
 - Calorific value ↑
 - Carbon content ↑
 - Moisture content ↓
- } → Quality of coke

Four varieties

- Anthracite: 80-90%
 - Bituminous: 60-80%
 - Lignite: 40-60%
 - Peat: <40%
- Coking coal (used in metallurgy)
- Also found in India
- Brown coal

Mostly found in India

Tertiary Coal



• **Reserve (top):** USA

India

• **Reserve (top):** Odisha > Jharkhand > Chhattisgarh

• **Production (top):** Chhattisgarh > Odisha

States

Mines

• **Jharkhand**

Jharia, Bokaro, Giridih, Karanpura

• **Chhattisgarh**

Hasdeo, Korba, Mand-Raigarh

• **Madhya Pradesh:** Singareni (coal mines)

• **Tamil Nadu**

Neyveli

• **Odisha**

Jharsuguda, Talcher, Ib valley

• **Tertiary Coal:** low-grade coal → Seen in J & K, Meghalaya

→ Peat is a tertiary coal

Oil and Natural Gas

→ Largest component of natural gas: Methane

India

• **Western offshore:** Mumbai High, Gujarat

• **Assam:** Digboi

Rare Earth Metals

• **Total:** 17 → 15 lanthanides + 2 (Yttrium, Scandium)

• **Topmost producer + reserve in world:** China

• **India (topmost):** Kerala (Producer); Andhra (Reserve)

→ Rich in Monazite sand, thorium

One Liners (MCQs)

- Main mineral constitutions of continental mass are: Silicon, Aluminium
- Hirakud Captive Thermal Power Plant is located in: Odisha
- The major resources of Gondwana coal, which are metallurgical coal are located in: Damodar valley
- Peat has low carbon content and high moisture content
- Reserves of silver found in: Odisha, Jharkhand, Andhra, Gujarat
- Amarkantak Plateau, Maikal hills and the plateau region of Bilaspur-katni are known for presence of: Bauxite
- H.V.J gas pipeline from Hazira in Gujarat to Jagdishpur in UP) passes through: Madhya Pradesh
 - H: Hazira in Gujarat
 - V: Vijaipur in MP
 - J: Jagdishpur in UP

India's 1st interstate gas pipeline, the project began in 1986, with the establishment of GAIL Ltd.

Guj-Raj-MP-UP-Haryana-Delhi
- Feldspar is a large group of rock-forming silicate minerals that constitutes more than 50% of the Earth's crust and is widely used in the glass and ceramic industries
- Jaduguda is know for Uranium deposits
 - In Singhbhum, Jharkhand
 - Also found in Cuddapah, Andhra